

CE 310 — Week 1 Assignment: Why Averages Lie

Due: Wed Sep 2, 9:00 AM — submit to D2L

Overview

This assignment builds directly on the Week 1 in-class exercise. You will extend your analysis beyond temperature and make a concrete design recommendation — the kind of decision a practicing engineer would write in a project report.

How to Submit

Read this before you submit. Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
Where to start	Download Wk01.06_Assignment_Answer_Sheet.xlsx from this week's D2L module. It is the template you fill in and submit — do not build your own workbook
File format	.xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF
Sheet name	Must be exactly Answers — do not rename or delete it
Where answers go	Column A of the Answers sheet labels every row — P1b_written, P2_written, and so on. Each question names its row; type into the shaded cell on that row
Column layout	Enter answers only in the labeled input cells for each problem — column position varies by section (see each section's own column headers). Do not insert, delete, or move columns

Requirement	Detail
Formulas	Type them as text — put an apostrophe before the =, e.g. '=COUNTIF(B2:B8761,">90")', so Excel stores the formula instead of running it. Write the formula itself, not a description of it — a missing or wrong formula loses the formula points on that question even when your answer is right. Screenshots are not accepted
Name and NetID	Type them in the Name and NetID cells at the top of the Answers sheet — that is what matches your work to your student record. A 5-point deduction applies if the NetID is blank.
File name	CE310_W1_Assignment_<NetID>.xlsx e.g. CE310_W1_Assignment_abc123.xlsx
Submission	Upload the .xlsx file to D2L → Assignments by 9:00 AM, Wed Sep 2

Problem 1 — Exceedance Hours (30 pts)

Using Week1_Tucson_Weather_2023.csv, answer the following.

1a. How many hours in 2023 did Tucson's temperature exceed each threshold below, and what share of the year is that?

Threshold (°F)	Hours exceeding	% of the year
85		
90		
105		

The 95 °F and 100 °F thresholds are not repeated here — you computed those in the in-class exercise as A6 and A7.

Use COUNTIF for the hour counts. On the Answer Sheet each threshold takes **two rows**: the hour count, then the percentage beneath it. Type the formula as text in **Col B** and the value in **Col C** — the same two columns Problem 1b and Problem 2 use further down the sheet.

The percentage row needs no formula of its own; it is the hour count above it divided by 8,760, and its Col B is marked with a dash.

1b. Using the ASHRAE 1% design dry-bulb temperature from your Week 1 in-class exercise workbook (the **88th-highest hourly temperature**, found with `=LARGE(B2:B8761, 88)`), calculate the peak **cooling load difference** between two design approaches for a 10,000 sq ft office building:

- **Approach A (wrong):** Size the system for the annual average temperature.
- **Approach B (correct):** Size the system for the ASHRAE 1% design temperature.

The simplified sensible cooling load formula is:

$$Q = U \times A \times \Delta T$$

where Q = cooling load (BTU/hr), $U = 0.5$ BTU/(hr·ft²·°F) (overall heat transfer coefficient), $A = 10,000$ ft² (floor area), and ΔT is the difference between the indoor setpoint (75 °F) and the outdoor design temperature.

On the sign of ΔT . Take the difference in whichever direction you were taught — indoor minus outdoor, or outdoor minus indoor. Both are marked correct for Q_A and Q_B , because this problem is about the *size* of the gap, not its direction. What matters is what the two magnitudes tell you: at an outdoor average near 72 °F against a 75 °F setpoint, Q_A is under 15,000 BTU/hr — the average day barely calls for cooling at all — while the design day calls for 165,000.

Calculate Q for both approaches and find the difference. Report **$Q_{\text{difference}} = |Q_B| - |Q_A|$** , a positive number either way. In 2–3 sentences, explain what it means in practice if the installed system can only handle Approach A's load on a peak summer day.

Enter your response in the shaded cell on row P1b_written of the Answers sheet.

1c. Using two numbers from your Week 1 in-class exercise workbook — the gap between the ASHRAE design temperature and the annual average (your answer to **B2**), and the hours above 100 °F (**A7**) — write 2–3 sentences explaining what a building occupant would experience during a peak summer day if the HVAC system was sized for the annual average temperature. Focus on comfort — when does the system fall short, and for how many hours of the year?

Enter your response in the shaded cell on row P1c_written of the Answers sheet.

Problem 2 — Seasonal Humidity and Ventilation Load (18 pts)

Your exercise computed the annual average relative humidity (B12) and hours above 60% RH (B13). Now compute the seasonal breakdown — this is new data not in your exercise.

2a. Calculate the average relative humidity for: - Winter: December, January, February (months 12, 1, 2) - Summer: June, July, August (months 6, 7, 8)

2b. Tucson's North American Monsoon runs July–September, driven by Gulf of California moisture, and ASHRAE 62.1-2022 requires that the outdoor air delivered to a space be sized from occupancy rate and floor area — air that has to be dehumidified before it reaches anyone. Using your two averages from 2a, explain in 3–4 sentences why sizing the latent load from the annual-average relative humidity rather than the monsoon peak underestimates it. Give one consequence for equipment capacity and one for the people in the building during the monsoon.

Enter your response in the shaded cell on row P2_written of the Answers sheet.

Problem 3 — Reflection (12 pts)

Answer in no more than **one paragraph (100–150 words)**:

A colleague argues: *"We should always design for the average case to keep costs reasonable — extreme events are rare and unpredictable anyway."*

Using at least one specific number from your analysis (an exceedance count, a peak value, or a pressure ratio), write a professional response that either agrees, disagrees, or qualifies this statement. Mention one real consequence (cost, safety, or occupant impact) of choosing the wrong design basis.

Enter your response in the shaded cell on row P3_written of the Answers sheet.

Grading Summary

Problem	Points
Problem 1a — 3 exceedance thresholds with formulas (3 pts each)	9
Problem 1b — Cooling load comparison + written explanation	12
Problem 1c — Written explanation of comfort impact	9
Problem 2 — Seasonal Humidity and Ventilation Load	18
Problem 3 — Reflection paragraph	12

Problem	Points
Total	60 pts